

EE-102 Lab 1 Report

Can Kurc
ID: 22502100

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Purpose of the Experiment

The purpose of this lab was to get used to using the lab equipment. We learned to use oscilloscopes, signal generators and breadboards. We tested 6 cases and introduced ourselves how everything worked.

1 Compensation

The first task was to compensate our oscilloscope probes. I set up my probe and oscilloscope and started compensation process. I had a passive probe with an attenuation factor of 10x. When I first plugged in my probe and checked the signal, it was messy with sloped tops (it was overcompensated). By adjusting the probe using the tool, I fixed my signal to be straight and well.

The compensation of probes means adjusting the capacitance between oscilloscopes to balance its electrical properties. It can be done by turning the screw on the probe with an adjusting rod [1].

I set up my probe and started compensation process. I had a passive probe with an attenuation factor of 10x. My probe was overcompensated at first (the tops had a slope) and I turned the screw to compensate it (The top of the signal was straight).



Figure 1.1: Me adjusting the probe

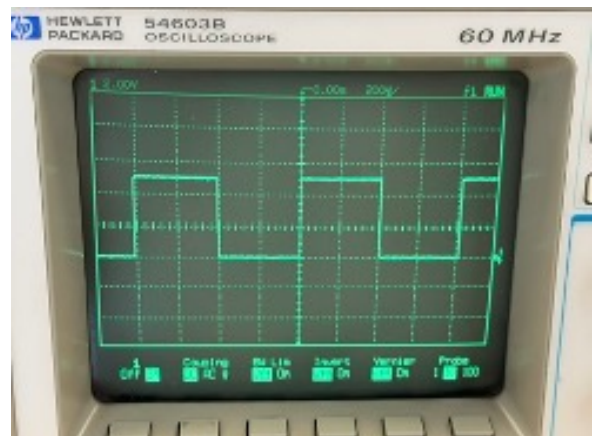


Figure 1.2: Compensated probe

2 Task 2

I applied a 5V peak to peak sinusoidal signal with a frequency of 1kHz which should include a DC component. I used the oscilloscope to monitor this signal (can be seen in figure 2.1). I first used positive edge triggering and saw the origin point occurring on the positive slope of the signal. When I applied negative edge triggering, I saw the origin point occurring on the negative slope of the signal.

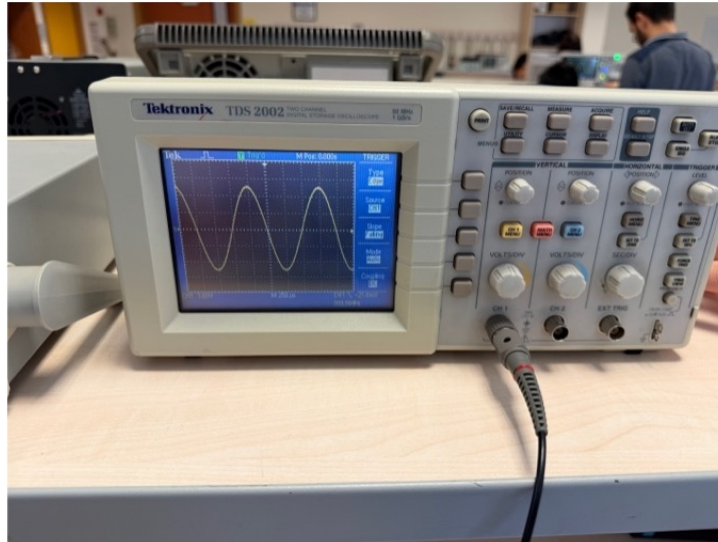


Figure 2.1: Sinusoidal signal

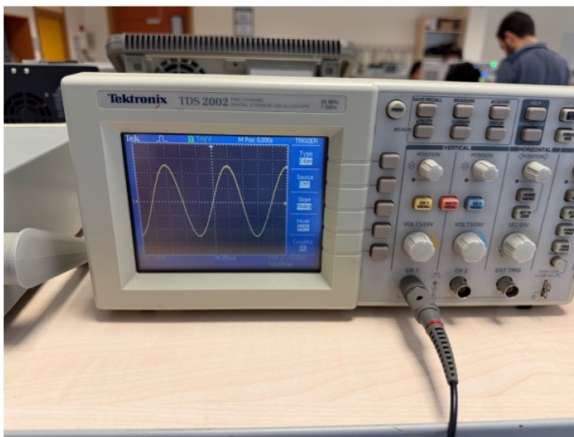


Figure 2.2: Positive edge triggering

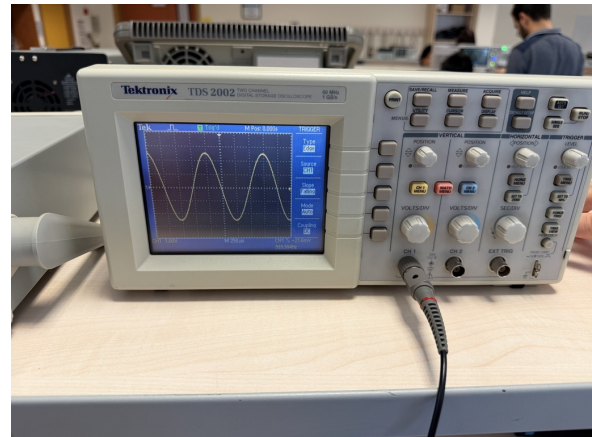


Figure 2.3: Negative edge triggering

3 Task 3

I applied a 1 Volt Peak to Peak triangular wave with a frequency of 2kHz. I used an oscilloscope to measure the wave and observed the effect of the trigger knob of the oscilloscope. The trigger knob stabilizes the display by setting a specific voltage threshold that the signal must cross before the oscilloscope begins drawing the trace. By forcing the sweep to start at the exact same point on the wave every time, it creates a steady,

“frozen” image rather than a drifting one. When I turned the trigger knob, the stable image of the triangular wave started moving making the wave extremely hard to analyze.

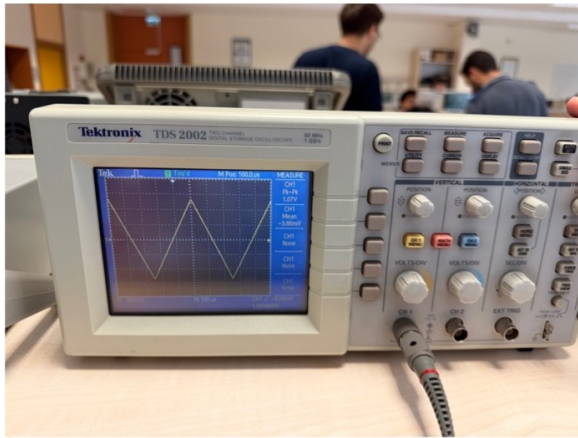


Figure 3.1: Triangular wave

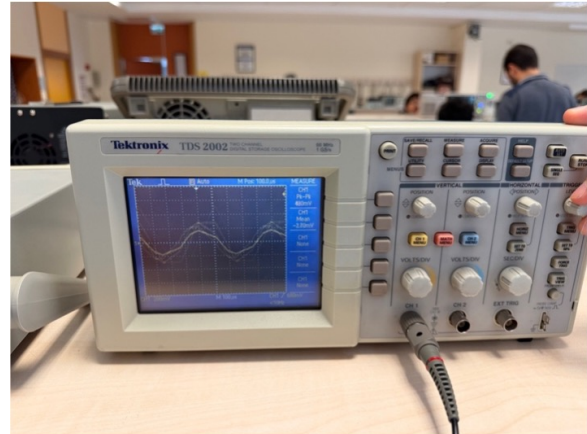


Figure 3.2: Wave after turning the trigger knob

4 Task 4

ADC (Analog-to-Digital Converter): Converts continuous “real-world” signals (like sound or voltage) into digital numbers (0s and 1s) that a computer can process.

DAC (Digital-to-Analog Converter): Does the reverse; it takes digital numbers and converts them back into a continuous physical signal (like playing music through a speaker).

In oscilloscopes, ADC are used. When a circuit is probed, the ADC samples the incoming analog voltage at very high speeds and converts those voltage levels into digital data points. This way, the oscilloscope can store the waveform in memory, perform calculations (like FFTs), and display stable graph on its screen.

I applied a 1 Volt Peak to Peak square wave with a frequency of 5kHz to the oscilloscope and observed that the sample mode provided a standard view of the wave, Peak Detect resulted in a thicker trace by capturing the extremes, and Average mode gave the clearest waveform by filtering out random noise.



Figure 4.1: Sample Mode

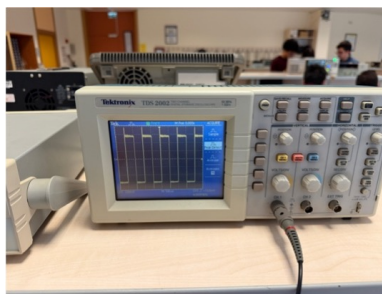


Figure 4.2: Peak Detect Mode

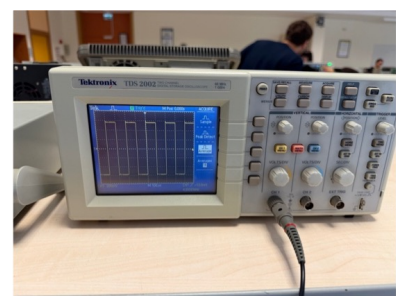


Figure 4.3: Average Mode

5 Task 5

I generated a sinusoidal signal with 2 Volts Peak to Peak and a frequency of 1 kHz, then applied a DC offset of 1 V to the wave. First, I used DC coupling on the oscilloscope and observed that the entire waveform was shifted upwards on the screen, oscillating above the ground reference because DC coupling displays the total voltage, including the 1 V offset. When I switched the setting to AC coupling, the waveform shifted vertically to be centered around the zero-volt line. This change occurred because AC coupling engages a capacitor that filters out the constant DC component of the signal, allowing only the alternating AC portion to be displayed, which is useful for analyzing the signal's shape without the offset.

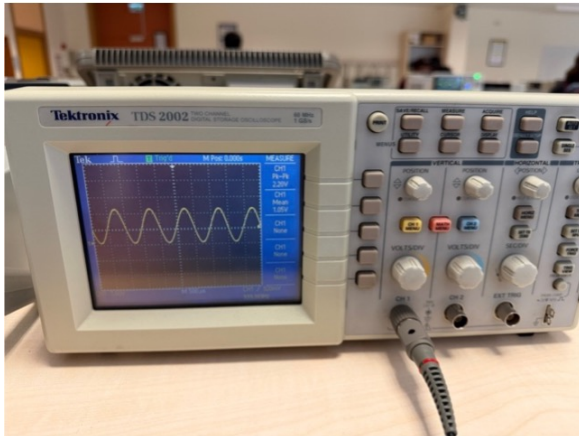


Figure 5.1: DC Coupling

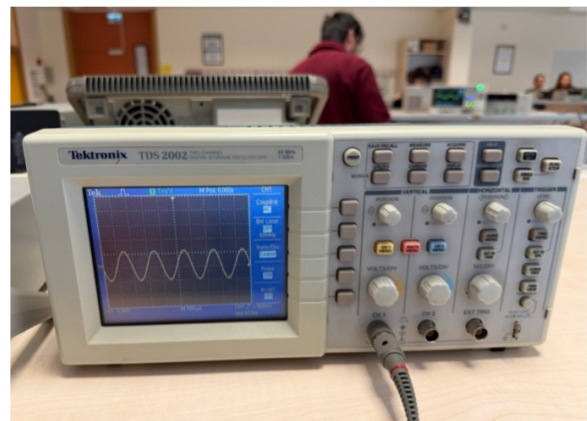


Figure 5.2: AC Coupling

6 Task 6

A breadboard is a construction base for prototyping electronics that allows for temporary circuit connections without soldering.

Structure: It consists of a plastic block with a grid of holes. Underneath, metal clips connect these holes in specific patterns.

Power Rails: The long vertical strips on the sides (usually marked with red and blue lines) are connected continuously down the column. These are used for power and ground distribution.

Component Rows: The horizontal rows in the center are connected in groups of five. Components inserted into the same row are electrically connected to each other, but isolated from adjacent rows.

I constructed an RC circuit on the breadboard using a $1\mu\text{F}$ capacitor and a $1\text{k}\Omega$ resistor as shown in the provided schematic.

Input (Point X): I connected the function generator to applying a 2 Vpp sinusoidal signal with a 0 V DC offset.

Output (Point Y): I probed the voltage across the resistor.

Grounding: The grounds of the function generator and both oscilloscope channels were

connected to the common ground rail on the breadboard to ensure a correct reference potential.

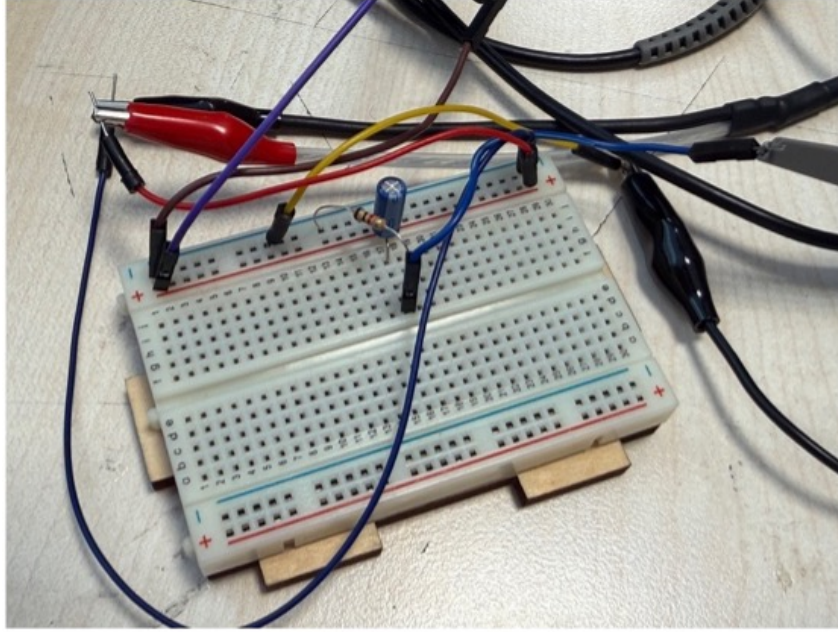


Figure 6.1: Breadboard design

When I applied a 1 kHz signal, I observed that the input (Channel 1, Yellow) and output (Channel 2, Blue) signals were similar in shape but not perfectly aligned. Using the oscilloscope's cursor function, I measured a specific time delay (Δt) between the two waveforms. The cursors indicated a time difference of $44.00 \mu s$.

Since the period (T) of a 1 kHz signal is $1000 \mu s$, I calculated the phase difference (ϕ) as follows:

$$\phi = \frac{\Delta t}{T} \times 360^\circ = \frac{44 \mu s}{1000 \mu s} \times 360^\circ \approx 15.8^\circ$$

This result confirms that at 1 kHz, the capacitor introduces a small but measurable phase shift, causing the voltage across the resistor to lead the input voltage slightly.

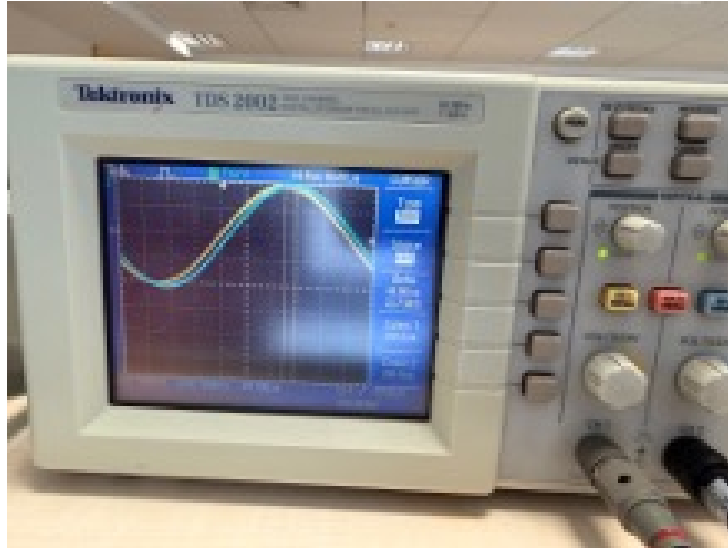


Figure 6.2: Waves 1 (1 kHz)

I then increased the frequency of the function generator to 100 kHz while keeping the amplitude constant.

Observation: At this higher frequency, the time delay between the input and output signals disappeared. The two waveforms overlapped almost perfectly on the display.

Analysis: This behavior is expected for the RC High-Pass filter we constructed. The capacitive reactance is calculated by $X_C = \frac{1}{2\pi fC}$.

At 1 kHz, the reactance is significant ($\approx 159\Omega$), creating the phase shift observed above.

At 100 kHz, the frequency (f) is much higher, causing the reactance (X_C) to drop close to zero. The capacitor effectively acts as a short circuit, allowing the signal to pass to the resistor with negligible impedance or phase delay.

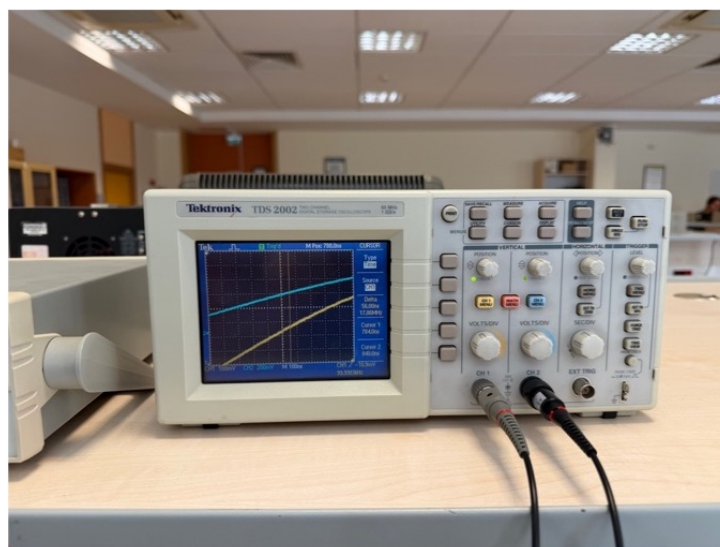


Figure 6.3: Waves 2 (100 kHz)

Conclusion

In this laboratory experiment, the fundamental operations of an oscilloscope and signal generator were successfully demonstrated. I verified that improper probe compensation can distort the signal display, making calibration a critical first step. The experiments highlighted the importance of triggering, showing that setting the correct edge and level is necessary for creating stable displays of repetitive signals.

I also observed that different acquisition modes serve specific purposes: 'Average' mode effectively filters random noise to create a clearer waveform, whereas 'Peak Detect' is better for capturing signal extremes. The distinction between AC and DC coupling was clearly demonstrated, with AC coupling proving useful for analyzing small AC signals riding on large DC offsets. Finally, the RC circuit experiment confirmed theoretical expectations regarding high-pass filters; we observed a measurable phase lead at lower frequencies (1 kHz) which diminished as the frequency increased (100 kHz). These tasks provided practical experience in signal measurement and circuit analysis essential for electrical engineering.

References

- [1] Digital Oscilloscope Principles. (n.d.). [PDF document].